12. APPLICATION OF 2ND ORDER ODE

Second order linear differential equations have a variety of applications in science and engineering. In this lesson, we explore one of them, spring-mass system (undamped and damped vibrations).

12.1 Undamped Vibrations

Objects vibrates with constant amplitude due to the absence of external resistance

We consider the motion of an object with mass m at the end of a spring that is either vertical (Figure 1) or horizontal on a level surface (Figure 2). By Hooke's Law, which says that if the spring is stretched (or compressed) x units from its natural length, then it exerts a force that is proportional to x:

$$restoring\ force = -kx$$

where k is a positive constant (called the **spring constant**).

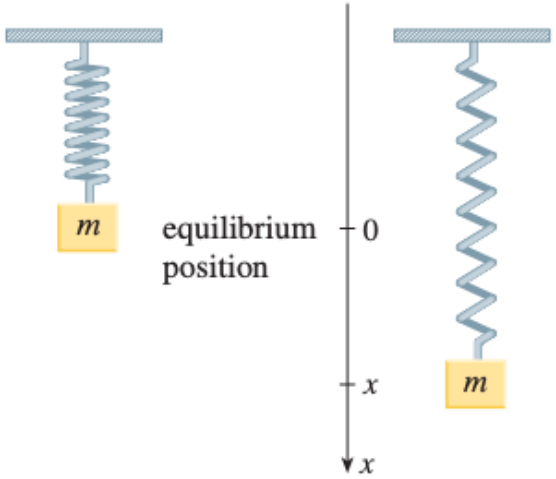


FIGURE 1

If we ignore any external resisting forces (due to air resistance or friction), then by Newton's Second Law, we have

$$ma = total\ force$$

$$m\frac{d^2x}{dt^2} = -kx$$

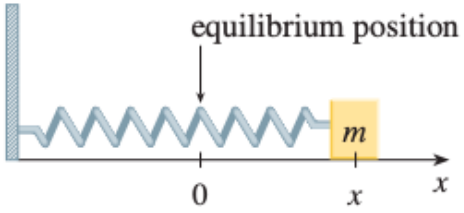


FIGURE 2

or

$$m\frac{d^2x}{dt^2} + kx = 0$$

This is a 2nd order linear differential equation, and its auxiliary equation is $mr^2 + k = 0$
The roots are

$$r = \pm\sqrt{\frac{k}{m}}\, i \quad \text{where} \quad \omega = \sqrt{\frac{k}{m}}$$

Thus, the general solution is

$$\boxed{x(t) = c_1 \cos \omega t + c_2 \sin \omega t}$$

$$A \sin \omega t \cos \delta + A \cos \omega t \sin \delta$$
$$A \sin \omega t \left(\frac{c_2}{A}\right) + A \cos \omega t \left(\frac{c_1}{A}\right)$$
$$c_2 \sin \omega t + c_1 \cos \omega t$$

which can also be written as an amplitude-phase angle form:

$x(t) = A\cos(\omega t - \delta)$

where $\omega = \sqrt{\frac{k}{m}}$ (frequency)

$A = \sqrt{c_1^2 + c_2^2}$ (amplitude)

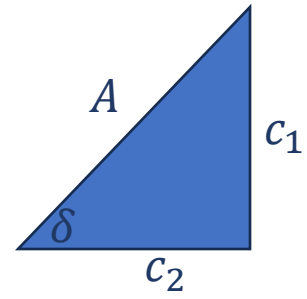
$\cos\delta = \frac{c_1}{A}$ $\sin\delta = \frac{c_2}{A}$ (δ is the phase angle)

$x(t) = A\sin(\omega t + \delta)$

where $\omega = \sqrt{\frac{k}{m}}$ (frequency)

$A = \sqrt{c_1^2 + c_2^2}$ (amplitude)

$\cos\delta = \frac{c_2}{A}$ $\sin\delta = \frac{c_1}{A}$ (δ is the phase angle)



This type of motion is called simple harmonic motion.

Example 12.1.1 A spring with a mass of $2\ kg$ has natural length $0.5\ m$. A force of $25.6\ N$ is required to maintain its stretched to a length of $0.7m$. If the spring is stretched to a length of $0.7m$ and then released with initial velocity 0, find the position of the mass at any time t.

Solution. Given that $m = 2\ kg$,

$$x = 0.7m - 0.5m = 0.2m$$
$$F = 25.6N$$
$$v_0 = 0$$

From Hooke's Law, the force required to stretch the spring is

$$25.6 = k(0.2)$$
$$k = \frac{25.6}{0.2}$$
$$= 128$$

By Newton's Second Law, we have

$$m\frac{d^2x}{dt^2} = -128x$$
$$2\frac{d^2x}{dt^2} + 128x = 0$$

The auxiliary equation is

$$2r^2 + 128 = 0 \quad \rightarrow r = \pm 8i$$

Thus, the general solution is

$$x(t) = c_1 \cos 8t + c_2 \sin 8t$$

Using the given initial condition $x(0) = 0.2,$ we have $0.2 = c_1$

Differentiating, we get

$$x'(t) = -8c_1 \sin 8t + 8c_2 \cos 8t$$

Using the given initial velocity $v(0) = 0,$ we have $0 = 8c_2 \rightarrow c_2 = 0$

$$\therefore x(t) = \frac{1}{5}\cos 8t$$

In amplitude phase-angle form:

$A = \sqrt{\left(\frac{1}{5}\right)^2 + 0} = \frac{1}{5}$

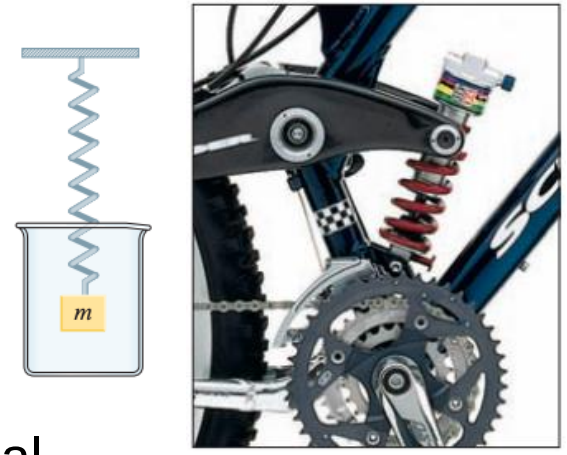
$\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{128}{2}} = 8$

$\cos\delta = \frac{1/5}{1/5} = 1$ and $\sin\delta = \frac{0}{1/5} = 0$ and $\tan\delta = 0$ → lies on the positive x-axis where $\delta = 0$

$$\therefore x(t) = A\cos(\omega t - \delta) = \frac{1}{5}\cos 8t$$

12.2 Damped Vibrations

Periodic vibrations of a body diminishing amplitude in the presence of a force



Damped vibrations occur when the spring motion is subject to external resisting forces:

- a frictional force (in the case of the horizontal spring)
- a damping force (in the case of a vertical spring moves to a fluid)
  - e.g. supplied by a shock absorber in a bicycle
  - e.g. car suspension system

In the study of mechanics, the damping force acting on a mass is proportional to the velocity of the mass

$$damping\ force = -b\frac{dx}{dt}$$

where b is a positive constant, called the **damping constant.** Then, by Newton's Second Law, we have

$$m\frac{d^2x}{dt^2} = restoring\ force + damping\ force$$

$$= -kx - b\frac{dx}{dt}$$

or

$$\boxed{m\frac{d^2x}{dt^2} + b\frac{dx}{dt} + kx = 0}$$

This is a 2nd order linear differential equation and its auxiliary equation is $mr^2 + br + k = 0$
The roots are

$$r_1 = \frac{-b + \sqrt{b^2 - 4mk}}{2m}, \qquad r_2 = \frac{-b - \sqrt{b^2 - 4mk}}{2m}$$

Now we need to discuss the 3 cases produced from these roots.

CASE 1: $b^2 - 4mk > 0$ (overdamping)
In this case, r_1 and r_2 are distinct real roots and the general solution is

$$x = c_1 e^{r_1 t} + c_2 e^{r_2 t}$$

Since b, m, and k are all positives, and $4mk < b^2$, thus we have $\sqrt{b^2 - 4mk} < b$. This makes

- the roots r_1 and r_2 must be both negative and thus $x \to 0$ as $t \to \infty$.
- damping constant is larger than spring constant.
- thus, the motion dies away with no oscillation.

CASE 2: $b^2 - 4mk = 0$ (critical damping)

In this case, r_1 and r_2 are equal roots where $r_1 = r_2 = -\frac{b}{2m}$ and the general solution is

$$x = c_1 e^{-\frac{b}{2m}t} + c_2 t e^{-\frac{b}{2m}t}$$

- the damping force balances the restoring force (the damping is sufficient to suppress vibrations).
- Any decrease in damping force leads to vibrations (oscillatory motion).

CASE 3: $b^2 - 4mk < 0$ (underdamping)

In this case, the roots are complex where $r = -\frac{b}{2m} \pm \omega i, \ \ \omega = \frac{\sqrt{4mk - b^2}}{2m}$ and the general solution is

$$x = e^{-\frac{b}{2m}t}(c_1 \cos \omega t + c_2 \sin \omega t)$$

- there are oscillations that are damped by the factor $e^{-\frac{b}{2m}t}$.
- Since $b > 0, m > 0$, we have $-\frac{b}{2m} < 0$. This leads to $e^{-\frac{b}{2m}t} \to 0$ as $t \to \infty$.
- thus, the motion decays to 0 as time increases.

Example 12.2.1 Suppose that the spring of Example 12.1.1 is immersed in a fluid with damping constant $b = 40$. Find the position of the mass at any time t if it starts from the equilibrium position and is given a push to start it with an initial velocity of $0.6\ m/s$.

In the study of vibrating spring with damping, we are led to an initial value problem

Solution. From example 12.1.1, we have $m = 2\ kg,\ k = 128$
We are given $x(0) = 0$ and $x'(0) = 0.6$

By Newton's Second Law, we have

$$m\frac{d^2x}{dt^2} = -128x - b\frac{dx}{dt}$$

$$2\frac{d^2x}{dt^2} + 40\frac{dx}{dt} + 128x = 0$$

The auxiliary equation is

$$r^2 + 20r + 64 = 0$$

$$(r + 4)(r + 16) = 0$$

$$r_1 = -4, r_2 = -16 \ \textbf{(case1 overdamping)}$$

Hence the general solution is

$$x(t) = c_1e^{-4t} + c_2e^{-16t}$$

$$x'(t) = -4c_1e^{-4t} - 16c_2e^{-16t}$$

Applying the initial conditions, we obtain:

$$x(0) = 0 \quad \Rightarrow \quad c_1 + c_2 = 0$$
$$x'(0) = 0.6 \quad \Rightarrow \quad -4c_1 - 16c_2 = 0.6$$

$$c_1 = 0.05 \text{ and } c_2 = -0.05$$

$$\therefore x(t) = 0.05e^{-4t} - 0.05e^{-16t}$$

Example 12.2.2 An object weighing $2.45\,kg$ stretches a spring $0.6125\,m$. Assuming that a damping force numerically equal to 2 times the instantaneous velocity acts on the system, determine the equation of motion if the object is initially released from the equilibrium position with an upward velocity of $3m/s$. (Use the acceleration due to gravity, $g = 9.8\,m/s^2$).

Solution. We are given $W = 2.45,\ x = 0.6125,$

Since weight is the gravitational force acting on an object

$$W = mg$$

$$m = \frac{W}{g} = \frac{2.45}{9.8} = \frac{1}{4}$$

Since a spring balance measures weight of a body, hence, restoring force is essentially the weight of an object. Thus, we have

Weight is measured by spring balance
Mass is measured by physical balance

$$2.45 = k(0.6125)$$

$$k = 4$$

By Newton's Second Law, we have

$$m\frac{d^2x}{dt^2} = -4x - 2\frac{dx}{dt}$$

$$\frac{1}{4}.\frac{d^2x}{dt^2} + 2\frac{dx}{dt} + 4x = 0$$

$$\frac{d^2x}{dt^2} + 8\frac{dx}{dt} + 16x = 0$$

The auxiliary equation is

$$r^2 + 8r + 16 = 0$$
$$(r+4)^2 = 0$$
$$r_1 = r_2 = -4$$ **(case2 critical damping)**

Hence the general solution is

$$x(t) = c_1 e^{-4t} + c_2 t e^{-4t}$$

$$x'(t) = -4c_1 e^{-4t} - 4c_2 t e^{-4t} + c_2 e^{-4t}$$
$$= -4c_1 e^{-4t} + (1 - 4t)c_2 e^{-4t}$$

Applying the initial conditions, we obtain:

$$x(0) = 0 \quad \Rightarrow \quad c_1 = 0$$
$$x'(0) = -3 \quad \Rightarrow \quad -4c_1 + c_2 = -3$$

$c_1 = 0$ and $c_2 = -3$

$$\therefore x(t) = -3te^{-4t}$$

Example 12.2.3 An object weighing $4.9\ kg$ is attached to a $3.5\ m$ spring. At equilibrium, the spring measures $4.48\ m$. If the object is initially released from rest at a point $2\ m$ above the equilibrium position, find the displacements $x(t)$ if it is further known that the surrounding medium offers a resistance numerically equal to the instantaneous velocity. (Use the acceleration due to gravity, $g = 9.8\ m/s^2$).

Solution. We are given $W = 4.9,\ x = 4.48 - 3.5 = 0.98,$

$$W = mg$$
$$m = \frac{4.9}{9.8} = \frac{1}{2}$$

Since a spring balance measures weight of a body, hence, restoring force is essentially the weight of an object. Thus, we have

$$4.9 = k(0.98)$$
$$k = 5$$

By Newton's Second Law, we have

$$m\frac{d^2x}{dt^2} = -5x - \frac{dx}{dt}$$
$$\frac{1}{2}.\frac{d^2x}{dt^2} + \frac{dx}{dt} + 5x = 0$$
$$\frac{d^2x}{dt^2} + 2\frac{dx}{dt} + 10x = 0$$

The auxiliary equation is

$$r^2 + 2r + 10 = 0$$

$$r_1 = -1 \pm 3i$$ **(case3 critical damping)**

Hence the general solution is

$$x(t) = e^{-t}(c_1 \cos 3t + c_2 \sin 3t)$$

$$x'(t) = e^{-t}(-3c_1 \sin 3t + 3c_2 \cos 3t) - e^{-t}(c_1 \cos 3t + c_2 \sin 3t)$$

Applying the initial conditions, we obtain:

$$x(0) = -2 \quad \Rightarrow \quad c_1 = -2$$

$$x'(0) = 0 \quad \Rightarrow \quad 3c_2 - c_1 = 0$$

$$c_1 = -2 \text{ and } c_2 = -\frac{2}{3}$$

$$\therefore x(t) = e^{-t}\left(-2\cos 3t - \frac{2}{3}\sin 3t\right)$$